STATE BOARD OF TECHNICAL EDUCATION — DIPLOMA COURSE NOTES

COURSE: DISCRETE MATHEMATICS (COURSE CODE: 3341)

SEMESTER III | DIPLOMA IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

MODULE 1: MATHEMATICAL LOGIC

Comprehensive Senior Lecturer Short Notes & Exam Preparation Guide

COURSE SYLLABUS ALIGNMENT & OBJECTIVES

Course Outcome CO1: Apply the basic concepts of mathematical logic to solve problems.
Duration: 16 Hours | **Cognitive Level:** Applying
Prescribed Textbooks: (1) Kenneth H. Rosen (7th Ed), (2) Ralph P. Grimaldi (5th Ed), (3) Thomas Koshy (2nd Ed).

1. Propositional Logic & Logical Operations

1.1 Concept of Propositional Logic (M1.01)

Definition (Proposition / Statement): A *proposition* (or statement) is a declarative sentence that is either **True (T)** or **False (F)**, but not both. The truth value of a proposition is denoted by T (or 1) for true and F (or 0) for false.

Sentence	Is it a Proposition?	Reason / Truth Value
Kerala is a state in India.	**YES**	Declarative sentence; Truth Value = **T**
2 + 3 = 7	**YES**	Declarative sentence; Truth Value = **F**
What time is it?	**NO**	Interrogative sentence (Question)
Solve this equation immediately!	**NO**	Imperative sentence (Command)
x + 5 = 12	**NO**	Open sentence; value depends on variable *x*

Atomic vs. Compound Propositions:

• **Atomic (Primitive) Proposition:** A simple proposition that cannot be broken down into simpler propositions (e.g., *p: 'It is raining'*).
• **Compound Proposition:** Formed by combining two or more atomic propositions using **logical connectives** (e.g., $p \wedge q$: *'It is raining AND it is cold'*).

1.2 Logical Connectives & Fundamental Truth Tables (M1.02)

Logical connectives combine propositions. Let *p* and *q* be propositions:

Connective	Symbol	Logical Name	Meaning / Condition for Truth
NOT	¬ (or ~)	Negation	Reverses truth value. True when proposition is false.
AND	$\wedge$	Conjunction	True **ONLY when BOTH** *p* and *q* are True.
OR	$\vee$	Disjunction (Inclusive)	True when **AT LEAST ONE** of *p, q* is True.
XOR	$\oplus$	Exclusive-OR	True when **EXACTLY ONE** of *p, q* is True (differs).
IF...THEN	$\rightarrow$	Implication / Conditional	False **ONLY when** *p* is True and *q* is False.

IFF	↔	Biconditional	True when *p* and *q* have the **SAME** truth value.

Master Truth Table for All Basic Logical Connectives:

p	q	¬p	p ∧ q	p ∨ q	p ⊕ q	p → q	p ↔ q
T	T	F	**T**	**T**	F	**T**	**T**
T	F	F	F	**T**	**T**	**F**	F
F	T	T	F	**T**	**T**	**T**	F
F	F	T	F	F	F	**T**	**T**

LECTURER'S DEEP DIVE ON IMPLICATION

Understanding Implication (p → q):

• *p* is called the **Hypothesis (Antecedent)**; *q* is called the **Conclusion (Consequent)**.

• The statement *p* → *q* is False **ONLY** when a True hypothesis leads to a False conclusion (Row 2).

• When *p* is False (Rows 3 & 4), the implication *p* → *q* is automatically **TRUE** regardless of *q*. This is called a **vacuous truth** in mathematics!

1.3 Step-by-Step Problem Solving: Truth Table Construction

Problem 1: Construct the truth table for the compound proposition *(p ∨ ¬q) → (p ∧ q)*.

Methodology: For *n* variables, construct a table with 2^n rows (here *n*=2, so 4 rows). Evaluate sub-expressions column by column from inside out.

Row	p	q	¬q	p ∨ ¬q (A)	p ∧ q (B)	A → B
1	T	T	F	T	T	**T**
2	T	F	T	T	F	**F**
3	F	T	F	F	F	**T**
4	F	F	T	T	F	**F**

2. Implication Variants, Truth Classifications & Normal Forms

2.1 Variants of Conditional Statements (M1.03)

Given an implication **p → q**, three related conditional statements can be formed:

Statement Name	Symbolic Form	English Translation	Logical Equivalence
Original Implication	**p → q**	If p, then q	Equivalence baseline
Converse	**q → p**	If q, then p	Equivalent to Inverse (**¬p → ¬q**)
Inverse	**¬p → ¬q**	If not p, then not q	Equivalent to Converse (**q → p**)
Contrapositive	**¬q → ¬p**	If not q, then not p	**STRICTLY EQUIVALENT to p → q**

KEY EXAMINATION FACT

CRITICAL LECTURER RULE:

An implication *p* → *q* and its **Contrapositive (¬q → ¬p)** ALWAYS have identical truth values in every row of a truth table! Hence, **p → q ≡ ¬q → ¬p**.

However, the Converse (*q* → *p*) and Inverse (¬*p* → ¬*q*) are NOT logically equivalent to the original implication!

Worked Example: Write the Converse, Inverse, and Contrapositive of the statement:
'If a polygon is a square (p), then it is a rectangle (q).'

- **Converse ($q \to p$):** If a polygon is a rectangle, then it is a square. (False)
- **Inverse ($\neg p \to \neg q$):** If a polygon is not a square, then it is not a rectangle. (False)
- **Contrapositive ($\neg q \to \neg p$):** If a polygon is not a rectangle, then it is not a square. (True)

2.2 Tautology, Contradiction, and Contingency (M1.03)

Based on their final truth values across all possible truth assignments, compound propositions are classified into three mutually exclusive categories:

Classification	Definition	Standard Example	Truth Table Characteristic
Tautology (T)	A compound proposition that is **ALWAYS TRUE** under all truth assignments.	$p \vee \neg p$	Final column contains **ONLY T**
Contradiction (F)	A compound proposition that is **ALWAYS FALSE** under all truth assignments.	$p \wedge \neg p$	Final column contains **ONLY F**
Contingency	A compound proposition that is **NEITHER** a tautology nor a contradiction.	$p \to q$	Final column has a **MIX of T and F**

Problem 2: Prove that $(p \wedge (p \to q)) \to q$ is a Tautology using a Truth Table.

p	q	$p \to q$	$p \wedge (p \to q)$	$[p \wedge (p \to q)] \to q$
T	T	T	T	**T**
T	F	F	F	**T**
F	T	T	F	**T**
F	F	T	F	**T**

Conclusion: Since all entries in the final column are T, $(p \wedge (p \to q)) \to q$ is a Tautology. (This is the basis of Modus Ponens!).

2.3 Normal Forms: DNF and CNF

In propositional logic, any compound proposition can be represented in standardized canonical forms:
- **Literal:** A variable p or its negation $\neg p$.
- **Disjunctive Normal Form (DNF / Sum-of-Products):** A disjunction (OR) of minterms, where each minterm is a conjunction (AND) of literals. Formed by selecting rows where the output is **TRUE (T)**.
- **Conjunctive Normal Form (CNF / Product-of-Sums):** A conjunction (AND) of maxterms, where each maxterm is a disjunction (OR) of literals. Formed by taking negated literals for rows where output is **FALSE (F)**.

3. Logical Equivalence & Duality Principle

3.1 Fundamental Laws of Logical Equivalences (M1.03)

Two propositions A and B are **logically equivalent ($A \equiv B$ or $A \Leftrightarrow B$)** if $A \leftrightarrow B$ is a tautology. The fundamental algebraic laws of logic (prescribed in Rosen & Grimaldi) allow simplification of logic circuits and algebraic proofs without truth tables.

Law Category	Equivalence Statement 1 (AND / OR)	Equivalence Statement 2 (Dual Form)
Identity Laws	$p \wedge T \equiv p$	$p \vee F \equiv p$

Domination Laws	$p \vee T \equiv T$	$p \wedge F \equiv F$
Idempotent Laws	$p \vee p \equiv p$	$p \wedge p \equiv p$
Double Negation Law	$\neg(\neg p) \equiv p$	—
Commutative Laws	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
Associative Laws	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
Distributive Laws	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
De Morgan's Laws	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$
Absorption Laws	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
Negation / Inverse Laws	$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$
Conditional Equivalences	$p \to q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

Problem 3 (Algebraic Proof without Truth Table): Prove that $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg q$.

Proof:

1. $\neg(p \vee (\neg p \wedge q)) \equiv \neg p \wedge \neg(\neg p \wedge q)$ [By De Morgan's Law]
2. $\equiv \neg p \wedge (\neg(\neg p) \vee \neg q)$ [By De Morgan's Law]
3. $\equiv \neg p \wedge (p \vee \neg q)$ [By Double Negation Law]
4. $\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$ [By Distributive Law]
5. $\equiv F \vee (\neg p \wedge \neg q)$ [By Negation Law ($\neg p \wedge p \equiv F$)]
6. $\equiv \neg p \wedge \neg q$ [By Identity Law ($F \vee A \equiv A$)] **Q.E.D.**

3.2 Duality Principle

Definition (Dual of a Proposition): Let s be a compound proposition containing only connectives $\wedge$, $\vee$, and $\neg$. The **dual of s**, denoted by **s***, is obtained by replacing:

- Every $\wedge$ with $\vee$, and every $\vee$ with $\wedge$.
- Every **T (Tautology)** with **F (Contradiction)**, and every **F** with **T**.
- *Note:* Negations ($\neg$) remain completely UNCHANGED!

Principle of Duality: If two compound propositions s and t are logically equivalent ($s \equiv t$), then their duals are also logically equivalent (**s*** ≡ **t***).

DUALITY WORKED EXAMPLES

Examples of Duality:

1. Statement: $(p \wedge q) \vee T \Rightarrow$ Dual: $(p \vee q) \wedge F$
2. Identity: $p \vee (p \wedge q) \equiv p \Rightarrow$ Dual Identity: $p \wedge (p \vee q) \equiv p$ (Absorption Law dual!)

4. Predicate Logic & Quantifiers

4.1 Predicates & Propositional Functions (M1.04)

Propositional logic cannot express statements like *'All prime numbers are odd except 2'*. Predicate logic extends propositional logic by introducing variables and predicates.

- **Predicate:** A statement of the form $P(x)$ where x is the subject variable and P is the predicate property assigned to x.
- **Domain (Universe of Discourse U):** The set of all possible values that variable x can take.
- **Well-Formed Formula (WFF):** A syntactically correct expression constructed using predicates, variables, connectives, and quantifiers.

4.2 Quantifiers (Universal & Existential)

Quantifiers express the extent to which a predicate applies over a domain *U*:

Quantifier Name	Symbolic Form	Meaning in English	When is it TRUE?	When is it FALSE?
Universal Quantifier	$\forall x\ P(x)$	For all x, P(x) For every x in U...	*P(x)* is True for **EVERY** $x \in U$.	There exists **AT LEAST ONE** *x* where *P(x)* is False (Counterexample).
Existential Quantifier	$\exists x\ P(x)$	There exists an x... For at least one x...	*P(x)* is True for **AT LEAST ONE** $x \in U$.	*P(x)* is False for **EVERY SINGLE** $x \in U$.

4.3 Negation of Quantified Statements (De Morgan's Laws for Quantifiers)

To negate a quantified statement, change the quantifier type and negate the predicate statement:

QUANTIFIER NEGATION RULES

1. $\neg(\forall x\ P(x)) \equiv \exists x\ \neg P(x)$ *('Not all x satisfy P' is equivalent to 'There exists an x that does NOT satisfy P')*
2. $\neg(\exists x\ P(x)) \equiv \forall x\ \neg P(x)$ *('There is no x satisfying P' is equivalent to 'Every x fails to satisfy P')*

Worked Example: Negate the statement: *'Every computer science student loves Discrete Mathematics.'*

- Let *C(x)*: 'x is a CS student', *M(x)*: 'x loves Discrete Mathematics'.
- Symbolic Form: $\forall x\ (C(x) \rightarrow M(x))$
- Negation: $\neg[\forall x\ (C(x) \rightarrow M(x))] \equiv \exists x\ \neg(C(x) \rightarrow M(x)) \equiv \exists x\ (C(x) \wedge \neg M(x))$
- English Translation: ***'There exists a computer science student who does NOT love Discrete Mathematics.'***

5. Rules of Inference & Natural Deduction

5.1 Logical Arguments & Validity (M1.05)

An **argument** in propositional logic is a sequence of propositions ending with a conclusion. The initial propositions are called **premises** ($P_1, P_2, ..., P_k$) and the final proposition is the **conclusion (C)**.
Definition of Valid Argument: An argument form with premises $P_1, P_2, ..., P_k$ and conclusion *C* is **VALID** if and only if the conditional statement $(P_1 \wedge P_2 \wedge ... \wedge P_k) \rightarrow C$ is a **Tautology**.

5.2 Rules of Inference for Propositional Logic

Rules of inference are standard valid argument forms used as building blocks in natural deduction proofs:

Rule of Inference	Premises	Conclusion	Name / Tautology Form
Modus Ponens (Mode of Affirming)	p $p \rightarrow q$	$\therefore q$	$[p \wedge (p \rightarrow q)] \rightarrow q$
Modus Tollens (Mode of Denying)	$\neg q$ $p \rightarrow q$	$\therefore \neg p$	$[\neg q \wedge (p \rightarrow q)] \rightarrow \neg p$
Hypothetical Syllogism	$p \rightarrow q$ $q \rightarrow r$	$\therefore p \rightarrow r$	$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$
Disjunctive Syllogism	$p \vee q$ $\neg p$	$\therefore q$	$[(p \vee q) \wedge \neg p] \rightarrow q$
Addition	p	$\therefore p \vee q$	$p \rightarrow (p \vee q)$
Simplification	$p \wedge q$	$\therefore p$	$(p \wedge q) \rightarrow p$

Conjunction	p q	∴ **p ∧ q**	[(p) ∧ (q)] → (p ∧ q)
Resolution	p ∨ q ¬p ∨ r	∴ **q ∨ r**	[(p ∨ q) ∧ (¬p ∨ r)] → (q ∨ r)

5.3 Rules of Inference for Quantified Statements

Rule Name	Premise	Conclusion	Condition / Restriction
Universal Instantiation (UI)	∀x P(x)	∴ **P(c)**	*c* is ANY element in the domain *U*.
Universal Generalization (UG)	P(c) for arbitrary c	∴ ∀**x P(x)**	*c* MUST be an **arbitrary** (not specific) element.
Existential Instantiation (EI)	∃x P(x)	∴ **P(c)**	*c* is a **SOME specific** element (must be new!).
Existential Generalization (EG)	P(c) for known c	∴ ∃**x P(x)**	Holds if *P(c)* is true for a known element *c*.

Problem 4 (Formal Natural Deduction Proof):
Given premises: *1. p → q*, *2. q → r*, *3. ¬r*, and *4. p ∨ s*.
Prove the conclusion: s.

Step	Proposition / Statement	Justification / Rule Used
1	p → q	Premise 1
2	q → r	Premise 2
3	p → r	**Hypothetical Syllogism** on Steps 1 and 2
4	¬r	Premise 3
5	¬p	**Modus Tollens** on Steps 3 and 4
6	p ∨ s	Premise 4
7	**s**	**Disjunctive Syllogism** on Steps 5 and 6 (Conclusion Proven!)

6. Methods of Proof & Resolution Principle

6.1 Standard Methods of Proof (M1.05)

Direct and indirect proof strategies for proving mathematical theorems of form $p \rightarrow q$:

Proof Method	Assumption Made	Goal to Demonstrate	Key Strategy / Logic
Direct Proof	Assume *p* is True	Show *q* is True	Use axioms, definitions, and logic directly.
Proof by Contraposition	Assume ¬*q* is True	Show ¬*p* is True	Uses logical equivalence $(p \rightarrow q) \equiv (\neg q \rightarrow \neg p)$.
Proof by Contradiction (Reductio ad Absurdum)	Assume ¬*p* (or $p \wedge \neg q$)	Derive a contradiction $r \wedge \neg r$ (F)	If assumption leads to nonsense, original statement must be True!
Proof by Cases	Assume $p = p_1 \vee p_2 \vee \ldots \vee p_k$	Show $p_i \rightarrow q$ for all *i*	Exhaustive evaluation of all distinct scenarios.

Classic Exam Proof (Proof by Contradiction): Prove that √2 is an Irrational Number.

Proof:
1. Assume for contradiction that **√2 is rational**. Then √2 can be written as a fraction *a / b* in lowest terms, where *a* and *b* are integers with **gcd(a, b) = 1**.
2. Squaring both sides: $2 = a^2 / b^2 \Rightarrow \mathbf{a^2 = 2b^2}$.

3. Since a^2 is a multiple of 2, a^2 is EVEN, which implies **a is EVEN**. Let $a = 2k$.
4. Substitute $a = 2k$ into equation: $(2k)^2 = 2b^2 \Rightarrow 4k^2 = 2b^2 \Rightarrow \mathbf{b^2 = 2k^2}$.
5. This implies b^2 is EVEN, so **b is also EVEN**.
6. If both a and b are even, they share a common factor of 2, contradicting our assumption that $gcd(a, b) = 1$!
7. ∴ Our initial assumption that √2 is rational is FALSE. Thus, **√2 is IRRATIONAL. Q.E.D.**

6.2 Resolution Principle & Resolution Refutation

The **Resolution Principle** (introduced by J.A. Robinson) is an algorithmic inference rule widely used in automated theorem proving and AI logic programming (e.g., Prolog).

• **Clause:** A disjunction (OR) of literals, e.g., $C_1 = p \vee \neg q \vee r$.

• **Resolution Rule:** Given two clauses containing complementary literals (p and $\neg p$):

Clause 1: **p ∨ C** Clause 2: **¬p ∨ D** ∴ **Resolvent: C ∨ D**

• **Resolution Refutation Proof Procedure (Proof by Contradiction):**

1. Convert all premises into Clause Form (CNF disjunctions).
2. Negate the conclusion to be proven and convert it into clause form.
3. Add the negated conclusion clause to the set of premise clauses.
4. Iteratively apply the resolution rule to pairs of clauses to generate new resolvent clauses.
5. If the **Empty Clause (■ or FALSE)** is derived, a contradiction is established, proving the original conclusion VALID!

Problem 5 (Resolution Refutation Proof):
Use the Resolution Principle to prove that the premises $p \rightarrow q$ and p entail conclusion q.

Step 1: Convert Premises to Clause Form:
• Premise 1: $p \rightarrow q \equiv \neg p \vee q \Rightarrow$ **Clause 1: (¬p ∨ q)**
• Premise 2: $p \Rightarrow$ **Clause 2: (p)**

Step 2: Negate the Conclusion:
• Conclusion to prove: $q \Rightarrow$ Negated Conclusion: **Clause 3: (¬q)**

Step 3: Perform Resolution Steps:
• Resolve Clause 1 $(\neg p \vee q)$ and Clause 2 (p) on literal p:
Clause 4 (Resolvent): (q)
• Resolve Clause 4 (q) and Clause 3 $(\neg q)$ on literal q:
Clause 5 (Resolvent): ■ (EMPTY CLAUSE / CONTRADICTION)

Conclusion: Since the empty clause ■ was derived, the argument is **VALID! Q.E.D.**

7. Senior Lecturer's Exam Revision & Problem Kit

7.1 Selected Polytechnic Model Question Set

Q1 (5 Marks): Construct the truth table for $(p \leftrightarrow q) \oplus (p \vee q)$ and state whether it is a tautology, contradiction, or contingency.

p	q	p ↔ q (A)	p ∨ q (B)	A ⊕ B
T	T	T	T	**F**
T	F	F	T	**T**

F	T	F	T	**T**
F	F	T	F	**T**

*Ans: Final column contains both T and F. Therefore, the compound proposition is a **CONTINGENCY**.*

Q2 (5 Marks): Express the statement *'For every real number x, if x > 2 then $x^2 > 4$'* in predicate logic and write its contrapositive and negation.

Solution:

• Let domain *U* be ■ (real numbers). Let *P(x)*: 'x > 2', *Q(x)*: '$x^2 > 4$'.

• **Symbolic Form:** $\forall x\ (P(x) \rightarrow Q(x))$

• **Contrapositive:** $\forall x\ (\neg Q(x) \rightarrow \neg P(x)) \Rightarrow$ *'For every real number x, if $x^2 \leq 4$ then $x \leq 2$.'*

• **Negation:** $\exists x\ (P(x) \wedge \neg Q(x)) \Rightarrow$ *'There exists a real number x such that $x > 2$ and $x^2 \leq 4$.'*

Q3 (5 Marks): Prove using Duality and De Morgan's laws that $\neg(p \wedge q) \equiv \neg p \vee \neg q$.

Ans: Refer to Section 3.1 & 3.2 for the step-by-step law applications.

HIGH-YIELD REVISION SUMMARY

POLYTECHNIC LECTURER'S FINAL CHECKLIST FOR MODULE 1 EXAM:

✔ Always state the number of rows (2^n) when constructing truth tables.
✔ Remember that $p \rightarrow q$ is False ONLY when T $\rightarrow$ F. In all other cases, it is True.
✔ Contrapositive $(\neg q \rightarrow \neg p)$ is logically equivalent to original implication $(p \rightarrow q)$.
✔ In algebraic proofs, explicitly mention the name of the law used at every single step.
✔ To negate a quantifier, swap $\forall \leftrightarrow \exists$ and negate the inner predicate statement.
✔ In Resolution Refutation proofs, ALWAYS negate the conclusion first!